INCH-POUND

MIL-G-50678C (AR)
AMENDMENT 2
28 December 1992
SUPERSEDING
AMENDMENT 1
24 June 1991

MILITARY SPECIFICATION

GATE, MULTIPLE, CIRCUITS: 9342754

This amendment forms a part of MIL-G-50678C (AR), dated 30 April 1990, and is approved for use by the U.S. Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1 and 2

2.1.1:
Delete the following specifications:

"MIL-S-45743 - Soldering, Manual Type, High Reliability Electrical and Electronic Equipment
MIL-S-46844 - Solder Bath Soldering of Printed Wiring Assemblies"

PAGE 2

* Delete the following military standard:

"MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies"

* 2.1.2: Add the following drawing:

"12950877 - Requirements for Soldering Electrical and Electronic Assemblies"

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3.2: Delete "e. Soldering electrical connections."

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* 3.3.1: Delete in its entirety and substitute the following:

"3.3.1 Vibration. The multiple gate circuits shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to the equivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrowband random-on-random data for equipment installed on the deck of the M113 armored personnel carrier."

* 3.3.3, 3.3.3.1 and 3.3.3.2: Delete in their entirety and substitute the following:

"3.3.3 Altitude operating. The multiple gate circuits shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg)."

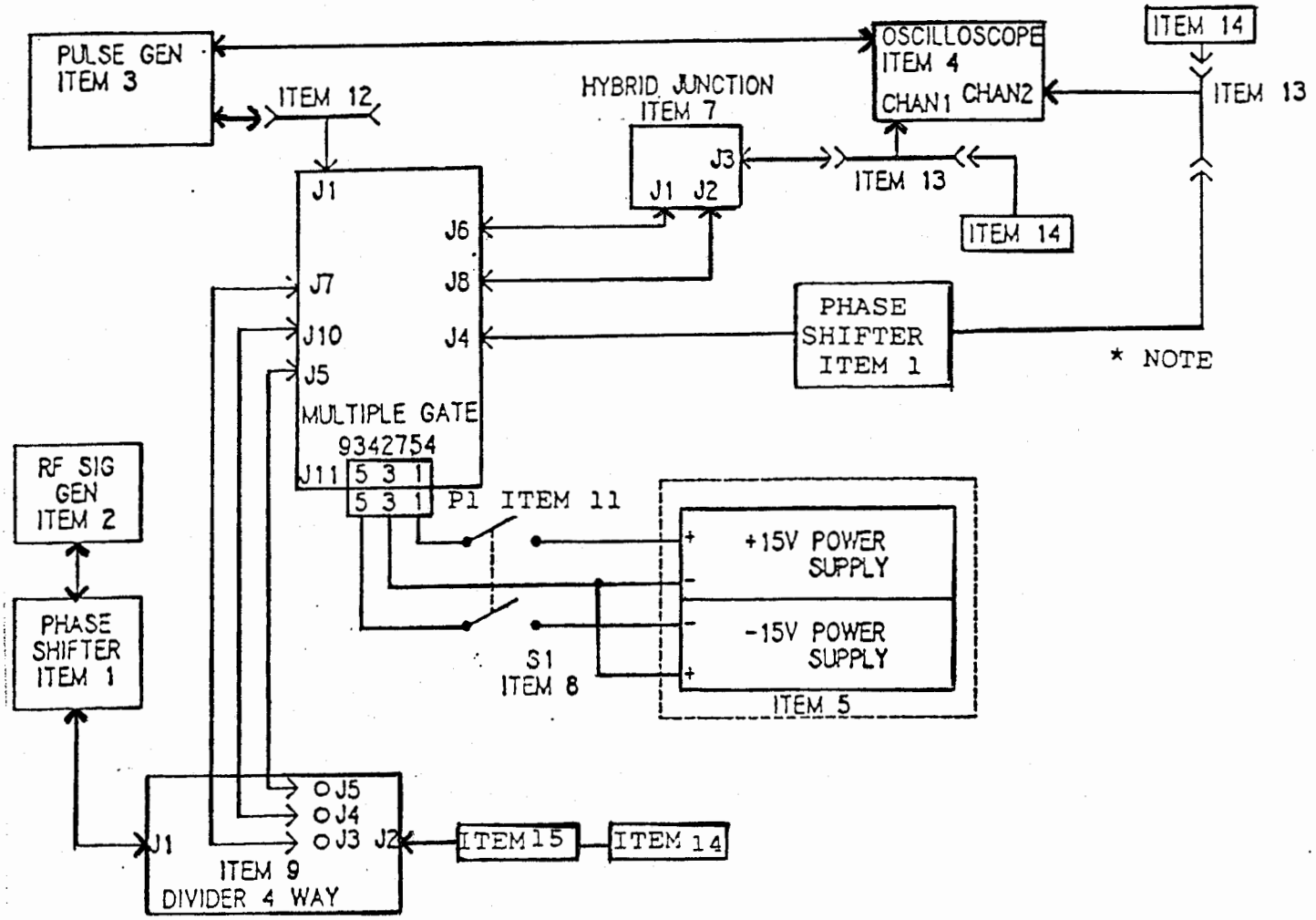
PAGE 6

Add new paragraph 3.5.2.3.4 as follows:

"3.5.2.3.4 Gate noise. With the multiple gate connected with the test setup as shown in Figure 5, a 60 ± 1% MHz signal and a monitored 2.8 V p-p signal per gate applied to J5, J7 and J10 of the unit under test, the amplitude of the algebraic sum of the Early Gage and Sum Gate signals during the Early Gate period shall be less than 1.0 V p-p. The amplitude of the algebraic sum of the Late Gate and inverted Sum Gage signals during the Late Gate period shall be less than 1.0 V p-p."

Add new Figure 5 as included page 6A.

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* NOTE:

Adjust Channel 2 CAble length as necessary to place Channel 1/
/Channel 2 Wave form relationship into phase shifter adjustment range.

FIGURE 5. Multiple gate circuit test setup.

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3.7: Delete in its entirety and substitute the following:

* "3.7 Soldering. All soldering shall be in accordance with the requirements of Note 2 of Drawing 12950877."

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4.6.2.3, Table III, Item 1:
Delete "1. Phase shifter Model PS-3-60, Merrimac or equal." and substitute "1. Phase shifter Model PS-3-60, Merrimac or equal (2)."

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* 4.7.1.1: Delete first sentence and substitute the following:

"4.7.1.1 Vibration. This test shall be performed as specified in Method 514, Category 8, Procedure I of MIL-STD-810 and paragraph 3.3.1 herein."

* 4.7.1.2: Delete "516.3" and substitute "516".

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* 4.7.1.3: Delete in its entirety and substitute the following:

"4.7.1.3 Altitude operating. This test shall be performed as specified in Procedure II, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3."

* 4.7.1.4.1: Delete "501.2" and substitute "501".

* 4.7.1.4.2: Delete "502.2" and substitute "502".

* 4.7.1.5: Delete "507.2" and substitute "507".

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Add new paragraph 4.7.3.4.4 as follows:

"4.7.3.4.4 Gate noise. Connect the unit under test to the test setup per Figure 5. Adjust channel 2 Phase Shifter so that the Early Gate on channel 1 is 180° out of phase with the Sum Gate on channel 2. Adjust variable gain on channel 2 for equal amplitude waveforms on channels 1 and 2. Place oscilloscope in channel 1 and 2 algebraic add function and readjust channel 2 phase shifter for Early Gate null. Resultant waveforms in the Early Gate shall be less than 1.0 V p-p. Place oscilloscope channel 2 in invert mode. Readjust amplitude and phase for Late Gate null. Resultant waveforms in the Late Gate shall be less than 1.0 V p-p."

Add new paragraph 4.8 as follows:

* "4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in paragraph 3.7."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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